

01_README

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{r setup, include=FALSE} knitr::opts_chunk$set(echo = TRUE)
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Module 01 — Quality, Normalization, Batch Correction & Ecological Profiling

Part of the CausoBiome Pipeline

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Version: 1.0

Objective

This module prepares high-quality, normalized, and batch-corrected microbial taxonomic and functional data for downstream biomarker discovery and causal inference in colorectal cancer (CRC). It integrates read-level quality control, taxonomic normalization, ARG/VF functional normalization, batch harmonization, and ecological diversity assessments.

Scripts Included in This Module

1. 01_mag_quality_metrics_analysis.py

Purpose: Assess the quality of MAGs (completeness, contamination, strain heterogeneity)

Inputs:

- SELECTED_BINS.csv — MAG quality metadata across cohorts

Outputs:

- Completeness & contamination KDE plots
- Strain heterogeneity violin plots
- Scatterplots and heatmaps of QC correlations
- Marker lineage distributions
- Regression analysis of N50 vs assembly features

2. 02_Taxon_counts_normalization.py

Purpose: Normalize species-level Bracken abundance using total reads per sample, then merge across cohorts.

Inputs:

- Bracken outputs (fractional read counts)
- Sample metadata (Merged_metadata_all.csv)

Outputs:

- Species_expression_matrix.csv — normalized species counts
- Species_metadata.csv — harmonized sample metadata

Functionality:

- Read scaling, per-sample aggregation, merging of cohort tables, metadata alignment

3. 03_Taxon_subset_pathogenic_samples.py

Purpose: Subset species matrix to include only samples positive for ARGs or VFs.

Inputs:

- `General_species_matrix.csv`
- Sample list of ARG/VF-positive samples (`vf_or_arg_samples`)

Outputs:

- `Species_subset_matrix.csv` — taxon profiles for ARG/VF-carrier samples

Context:

Used prior to batch correction to harmonize inputs for functional ecology.

4. 04_taxon_batch_correction.R

Purpose: Perform batch correction on the taxonomic matrix using ComBat.

Inputs:

- `Species_subset_matrix.csv`
- `crc_meta.csv` — metadata with batch covariates

Outputs:

- `combat_corrected_matrix.csv` — corrected counts
- `hellinger_transformed_combat.csv` — Hellinger-transformed version
- PCA, UMAP, and PERMANOVA outputs pre/post correction

Key Methods:

- Cramér's V for confounding assessment
- PCA + UMAP visualization
- PERMANOVA (bray) to validate correction

5. 05_taxon_ecology.R

Purpose: Assess ecological diversity (alpha, beta) on batch-corrected taxonomic profiles.

Inputs:

- `hellinger_transformed_combat.csv`
- `Aligned_metadata_Taxonomy.csv`

Outputs:

- Alpha diversity statistics and plots (Shannon, Simpson)
- Beta diversity: NMDS, ANOSIM, betadisper
- Stressplots and Cliff's Delta for pairwise contrasts

Figures:

- `Alpha_Diversity_Panel.png`, `NMDS_Only.png`, `ANOSIM_species.tiff`
- `beta_diversity_summary.csv`, `Stressplot_Only.png`

6. 06_arg_vf_normalization.py

Purpose: Normalize CARD and VFDB gene hits per sample and apply Hellinger transformation.

Inputs:

- `Merged_CARD.csv` and `Merged_VFDB.csv`

Outputs:

- CRC_CARD_expression_matrix_hellinger.csv
- CRC_VFDB_expression_matrix_hellinger.csv

Steps:

- Presence-absence counting
- Relative abundance calculation
- Gene-level aggregation
- Transformation for ecological analysis

7. 07_arg_vf_batch_correction.R

Purpose: Apply batch correction to merged ARG + VF matrices using `limma::removeBatchEffect`.

Inputs:

- Merged Hellinger matrices from step 6
- Metadata (CRC_vf_arg_meta_main.csv)

Outputs:

- Combined_ARG_VFDB_batch_corrected.csv
- Final matrix with negatives zeroed: combined_ARG_VFDB_final_DATA.csv
- UMAP and PERMANOVA diagnostics before/after correction

Batch variables: Project, Center, Instrument

8. 08_arg_vf_ecology.R

Purpose: Run ecological analyses (alpha/beta diversity, dispersion, NMDS, Procrustes) on batch-corrected ARG/VF data.

Inputs:

- combined_ARG_VFDB_final_DATA.csv
- Metadata_Aligned_to_CountMatrix.csv

Outputs:

- Alpha diversity boxplots + LM model predictions
- adonis_ARG_adjusted.csv, adonis_VF_adjusted.csv
- NMDS_ARG_plot.png, NMDS_VF_plot.png
- Procrustes correlation: procrustes_statistics.csv
- Null model tests: oecosimu_ARG_results.txt, oecosimu_VF_results.txt

Dependencies

Python (Scripts 1, 2, 3, 6):

- pandas, seaborn, matplotlib, statsmodels, numpy

R (Scripts 4, 5, 7, 8):

- vegan, sva, limma, ggplot2, ggpubr, patchwork, effsize, ggeffects, umap, car, broom, dplyr

Outputs Summary

Script	Output Type	Example Files
01	QC Plots	CRC_MAGS_completeness_distribution.png
02	Matrix	Species_expression_matrix.csv
03	Subset	Species_subset_matrix.csv

Script	Output Type	Example Files
04	Corrected Data + PCA/UMAP	combat_corrected_matrix.csv, UMAP_Before_After_Correction.tiff
05	Ecology Stats	adonis_adjusted.csv, beta_diversity_summary.csv
06	ARG/VF matrices	CRC_CARD_expression_matrix_hellinger.csv
07	Batch-Corrected ARG/VF	combined_ARG_VFDB_final_DATA.csv
08	Ecology Visuals	NMDS_ARG_plot.png, ARG_Beta_dispersion.png